Supplementary Material: A Validated Leakage Audit for Wearable Physiological Regression, with Application to PPG-Based Glucose and Blood-Pressure Estimation

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S1. TRIPOD+AI checklist

Items of the TRIPOD+AI statement [1] and where each is addressed (main paper: Sec. I–VI; supplement: S2–S11). Items that do not apply to an evaluation of public data are marked n/a.

Item	Requirement	Where
1	Title identifies study as developing/evaluating prediction models	Title, Abstract
2	Abstract: objectives, data, methods, results, conclusions	Structured abstract
3a–b	Background, rationale, objectives	Sec. I
4	Data sources, dates, setting	Sec. IV-A; S3, S7
5a	Participants: eligibility, recruitment	Sec. IV-A, S3 (as reported by dataset authors)
5b–c	Treatment / care pathway	n/a
6	Data preparation: cleaning, windowing, missing data	Sec. IV-A; S2; loaders in code
7	Outcome definition and measurement	Sec. IV-A; S3 (fingerstick meter; Dexcom G6; cuff BP)
8	Predictors: definition and measurement	Sec. IV-A; S2
9	Sample size justification	Sec. III, IV-D (power of the decision rule); S5, S9
10	Missing-data handling	S2 (gap-free windows only)
11	Analytical methods: model type, hyper-parameters, selection	Sec. IV-A; S2
12a	Evaluation: splits, protocol, leakage control	Sec. II-C, II-D; Table I; frozen folds in code
12b	Model performance measures	Sec. II-E
12c	Model comparison and null models	Sec. II-E; S11
12d	Uncertainty (CIs, tests)	Sec. II-E; Tables II–V; S6–S9, S11
12e	Fairness / subgroup analysis	Per-subject results (Fig. S3b); sex not analysed (n too small)
13	Class imbalance / outcome distribution	S3; glucose ranges and hypo-/hyperglycaemic counts
14	Open science: code, data, protocol	Title footnote; README (incl. encoder weight sources and licences)
15	Patient and public involvement	n/a (public data)
16–17	Participant flow, characteristics	Sec. IV-A; S3
18a–f	Model performance, comparisons, uncertainty, subgroups	Sec. IV-B–IV-D; S6–S11
19	Model updating	n/a
20	Interpretation; comparison with prior work	Sec. V; Table S9
21	Limitations	Sec. V
22	Usability: inputs, outputs, intended use	S2 (intended use; nulls per use case in Sec. II-E)
23	Funding, conflicts	To be completed by the author

S2. Model information sheet

Field	Value
Task	Regression of blood / interstitial glucose (mg/dL) from a 5-s PPG window (networks, classical) or 10–30-s segments (frozen encoders); systolic and diastolic blood pressure (mmHg) from 2.1-s PPG-BP segments
Intended use	Benchmark only; no clinical use
Inputs	320 samples at 64 Hz, 0.5–8 Hz bandpassed, z-scored per window (variants in Table S15)
Outputs	One glucose value per window; averaged per reference measurement
Training data	Kermani–Esmaeili recordings (LOSO folds); BIG IDEAs training periods; STUBA ePPG (LOSO); PPG-BP (5-fold subject-wise and random-segment folds)
Hyper-parameters	AdamW, lr 10^{-3}, wd 10^{-4}, one-cycle, 40/15 epochs, batch 16/64, grad-clip 1, smooth-L1 on standardised target
Seeds	Kermani–Esmaeili networks: 3; STUBA 1-D CNN: 3, other STUBA networks one; BIG IDEAs residual 1-D CNN: 3, other BIG IDEAs rows one; encoder probes and classical models are deterministic (one run)
Null models	Train mean; subject train mean; time of day; persistence
Leakage controls	Named protocols with asserted guarantees; frozen folds; guard gap; label-construction detector, whole-column and block scan
Known failure modes	Predicts near the (subject) mean; no event sensitivity
Pretrained encoders	PaPaGei-S (5.8 M, 125 Hz, 10-s), Pulse-PPG (28.5 M, 50 Hz, 10–30-s), AnyPPG (4.0 M, 125 Hz, 10-s); frozen released weights loaded weights-only; 0.5–12 Hz Chebyshev-II, resample, z-score; 512-d embeddings averaged per reference; RidgeCV ($\alpha \in [10^{-2}, 10^{4}]$) on training-fold-standardised features
Licences	Data: Kermani–Esmaeili CC BY 4.0 (Mendeley Data); BIG IDEAs PhysioNet ODC-By; STUBA ePPG CC0; PPG-BP CC0 (figshare). Encoders: PaPaGei code BSD-3-Clause, weights Zenodo 13983110 (no licence stated); Pulse-PPG MIT (code and weights, Zenodo 17345536); AnyPPG checkpoint from GitHub PKUDigitalHealth/AnyPPG (no licence file)

S3. Cohort characteristics

	Kermani–Esmaeili	**BIG IDEAs**	**STUBA ePPG**	**PPG-BP**
Subjects	23	16	24	219
Male / female	15 / 8	7 / 9	20 / 4	see [2]
Age (years)	22–61	35–65 (inclusion criterion)	22–48	21–86
Diabetes	not stated	HbA1c 5.3–6.4%	3 T1DM	not an outcome
Reference	Accu-Chek Active	Dexcom G6 CGM	Beurer GL49	cuff (Omron HEM-7201)
Reference measurements	67	28867	125	219 (657 segments)
Glucose range (mg/dL)	88–183	40–261	40–304	–
< 70 / > 180 mg/dL	0 / 2	186 / 431	34 / 18	–
Sensor	finger, 550 nm, 2175 Hz	wrist E4 BVP, 64 Hz	finger, 530/655/940 nm, 100 Hz	finger, 1000 Hz, 2.1-s segments

S4. Validation of the label-construction detector

Synthetic label columns of 20 subjects, 200 rows per reference value; a corrupted subject has a contiguous fraction of its rows replaced by the cross-join pattern (its k reference values cycled row by row). Thirty replicates per condition. The detector is `detect_label_cycling` (whole column) and `detect_label_cycling_local` (block scan, $b = 256$, stride 128, $\tau = 0.98$, $\ell_{\max} = 32$) in `ppgbench.integrity`; the experiment is `validate_detectors.py`. Residual misses at $k = 2$ are draws in which a subject's two reference values coincide, so that the column contains no cycle to detect.

TABLE S3

Cross-join detector on synthetic label columns: subject-level sensitivity (%) by cycle length k and fraction of a corrupted subject's rows that cycle, pooled over 10–100 % corrupted subjects. Specificity: 100 % (whole-column) and 100 % (block scan) in every condition.

	Whole-column			**Block scan**		
k	25 %	50 %	100 %	25 %	50 %	100 %
2	99	98	98	99	98	98
3	100	100	100	100	100	100
5	100	100	100	100	100	100
7	0	100	100	100	100	100
10	0	4	100	100	100	100

Real negative control, Kermani–Esmaeili: 0 / 0 of 23 subjects flagged (whole-column / block)
Real negative control, STUBA ePPG: 0 / 0 of 24 subjects flagged (whole-column / block)
Real negative control, BIG IDEAs: 0 / 0 of 16 subjects flagged (whole-column / block)
Real positive control, Kaggle derivative: 21 / 21 of 21 patients with > 1 reference value flagged

S5. Validation of the null-anchored decision rule

Each synthetic cohort has 16 subjects $\times$ 12 records $\times$ three overlapping 5-s windows generated by a pulse model with heart-rate variability, respiration and noise. Label = subject offset ($\mathcal{N}(120, 20^2)$, not encoded in the signal) $+ g\,(\mathrm{HR} - 70)$ $+ \mathcal{N}(0, 3^2)$, with within-subject HR SD 8 bpm. Each subject has a label-independent pulse-shape fingerprint (second-harmonic amplitude $\sim U(0.15, 0.75)$, notch amplitude $\sim U(0.05, 0.45)$). Models use the nine hand-crafted features of the main paper; predictions are averaged per record and scored with 2 000 subject-cluster bootstrap draws.

TABLE S4

Null-anchored decision rule on synthetic cohorts (200 per condition): percentage of cohorts in which the model is declared better than the train-mean null (95 % subject-cluster interval of ΔMAE entirely below zero), and the mean ΔMAE (mg/dL). g: true learnable signal in mg/dL per bpm; at $g = 0$ every claim is false (nominal one-sided rate 2.5 %).

Protocol	**Model**	$g = 0$	$g = 1$	$g = 2$	$g = 3$	$g = 4$
subject-wise (honest)	ridge	1.0 % (+1.7)	5.5 % (+0.1)	31.5 % (-4.4)	80.0 % (-11.3)	96.5 % (-18.7)
subject-wise (honest)	GBM	1.0 % (+3.1)	2.0 % (+1.7)	20.5 % (-2.9)	69.0 % (-10.3)	93.0 % (-17.3)
random windows (leaky)	ridge	19.0 % (-1.9)	53.0 % (-3.9)	95.5 % (-8.6)	100.0 % (-15.3)	100.0 % (-22.3)
random windows (leaky)	GBM	100.0 % (-9.3)	100.0 % (-11.6)	100.0 % (-16.5)	100.0 % (-23.3)	100.0 % (-30.1)

S6. PPG-BP: full results

S7. VitalDB arterial-line blood pressure: full results

Cohort: a seeded random sample of 600 of the 3 459 VitalDB cases with both `SNUADC/PLETH` and `SNUADC/ART` [3], read at 100 Hz with the official client; 10-s segments every 120 s from 10 min after the start to 10 min before the end; quality

TABLE S5
PPG-BP, ALL ROWS. MAE AND RMSE IN MMHG; NETWORKS MEAN ± STD OVER SEEDS; Δ: PAIRED MAE DIFFERENCE AGAINST THE TRAIN-MEAN NULL WITH ITS 95 % SUBJECT-CLUSTER INTERVAL (MOST CONSERVATIVE INTERVAL OVER SEEDS).

Target	**Split**	**Model**	**MAE**	**RMSE**	r	Δ [95% CI]
SBP	random segments	Train mean (null)	16.3	20.4	-0.16	–
SBP	random segments	1-NN copy probe	16.9	21.8	0.22	+0.6 [-0.88, +2.09]
SBP	random segments	Ridge, demographics	13.7	18.0	0.47	-2.6 [-3.65, -1.51]
SBP	random segments	Ridge, PPG features	15.0	19.1	0.35	-1.3 [-1.84, -0.72]
SBP	random segments	Ridge, PPG + demogr.	13.6	17.5	0.51	-2.7 [-3.71, -1.65]
SBP	random segments	GBM, PPG features	13.0	17.3	0.53	-3.3 [-4.40, -2.16]
SBP	random segments	GBM, PPG + demogr.	9.4	11.9	0.82	-6.8 [-8.28, -5.53]
SBP	random segments	1-D CNN	12.2±0.2	15.7±0.2	0.64±0.01	-4.1 [-5.14, -2.55]
SBP	subject-wise	Train mean (null)	16.2	20.4	-0.12	–
SBP	subject-wise	1-NN copy probe	17.1	22.4	0.17	+0.8 [-0.71, +2.38]
SBP	subject-wise	Ridge, demographics	13.9	18.3	0.44	-2.4 [-3.47, -1.30]
SBP	subject-wise	Ridge, PPG features	15.1	19.3	0.31	-1.1 [-1.73, -0.55]
SBP	subject-wise	Ridge, PPG + demogr.	13.8	18.0	0.47	-2.4 [-3.48, -1.35]
SBP	subject-wise	GBM, PPG features	14.5	18.9	0.38	-1.7 [-2.82, -0.56]
SBP	subject-wise	GBM, PPG + demogr.	14.6	18.6	0.44	-1.7 [-3.13, -0.30]
SBP	subject-wise	1-D CNN	13.4±0.1	17.2±0.1	0.54±0.01	-2.9 [-4.21, -1.36]
DBP	random segments	Train mean (null)	8.7	11.1	-0.06	–
DBP	random segments	1-NN copy probe	9.9	12.6	0.12	+1.2 [+0.33, +2.07]
DBP	random segments	Ridge, demographics	8.4	10.7	0.26	-0.3 [-0.65, +0.03]
DBP	random segments	Ridge, PPG features	8.6	10.9	0.17	-0.1 [-0.25, +0.09]
DBP	random segments	Ridge, PPG + demogr.	8.1	10.4	0.35	-0.6 [-1.01, -0.18]
DBP	random segments	GBM, PPG features	8.1	10.4	0.36	-0.6 [-1.17, -0.04]
DBP	random segments	GBM, PPG + demogr.	5.5	7.2	0.78	-3.2 [-3.89, -2.48]
DBP	random segments	1-D CNN	7.2±0.1	9.3±0.2	0.55±0.03	-1.5 [-2.02, -0.72]
DBP	subject-wise	Train mean (null)	8.7	11.1	-0.05	–
DBP	subject-wise	1-NN copy probe	10.2	13.1	-0.00	+1.4 [+0.62, +2.23]
DBP	subject-wise	Ridge, demographics	8.5	10.8	0.23	-0.2 [-0.53, +0.16]
DBP	subject-wise	Ridge, PPG features	8.7	11.1	0.07	+0.0 [-0.16, +0.19]
DBP	subject-wise	Ridge, PPG + demogr.	8.4	10.6	0.30	-0.4 [-0.81, +0.07]
DBP	subject-wise	GBM, PPG features	8.6	11.0	0.21	-0.1 [-0.64, +0.43]
DBP	subject-wise	GBM, PPG + demogr.	8.2	10.4	0.38	-0.6 [-1.25, +0.12]
DBP	subject-wise	1-D CNN	7.9±0.1	10.2±0.1	0.42±0.01	-0.8 [-1.39, -0.05]

control and beat-wise labels as in `scripts/fetch_vitaldb.py`; at most 40 segments per case. Segments: 21412; cases: 558; patients: 556. Labels are the median beat systolic peak and the median inter-beat trough of the arterial trace over the 10-s segment. Quality control has no arterial-line damping check and no PPG–arterial lag check; missing demographics were replaced by the cohort mean. The personal protocols train on the first half of every case's segments and test on the second half; their null is each case's training-half mean. In the absolute variant models predict pressure directly (they receive no calibration input); in the residual variant they learn the deviation from each case's training mean and add it back, so they receive the same calibration as the null. All VitalDB models were run with one seed.

S8. RE-AUDIT OF A PUBLISHED CLAIM: EVERY SEED

S9. ADDITIONAL VALIDATION

Stress tests of the cross-join detector, validation of the copy probe, the decision rule's false-claim rate on 1 000 additional no-signal cohorts, the power of the rule at glucose-cohort sizes, and seed-ensemble decisions.

S10. PPG → GLUCOSE: DETAILED RESULTS

Tables and figures that support Sec. IV-B of the main paper. The fusion network is shown in Fig. S1; published results on the Kermani–Esmaeili recordings are in Table S9; the full leakage ladder is in Table S10.

TABLE S6

VitalDB arterial-line blood pressure, all rows. MAE, mean error (ME) and SD of error in mmHg (AAMI/ISO 81060-2 limits: $|\text{ME}| \leq 5$, $\text{SD} \leq 8$); r: pooled correlation; r_w: median within-case correlation; Δ: paired MAE difference against the protocol's null with its 95 % patient-cluster interval.

Target	Split	Model	MAE	ME	SD	r	r_w	Δ [95% CI]
SBP	random segments	Null (train mean; case mean if personal)	16.2	0.0	20.4	-0.01	-0.00	–
SBP	random segments	1-NN copy probe	17.5	-0.9	22.3	0.19	0.18	+1.29 [+1.01, +1.55]
SBP	random segments	Ridge, demographics	16.2	-0.0	20.4	0.05	-0.02	-0.02 [-0.08, +0.03]
SBP	random segments	Ridge, PPG features	15.3	0.0	19.3	0.33	0.53	-0.95 [-1.16, -0.74]
SBP	random segments	Ridge, PPG + demogr.	15.2	0.0	19.2	0.34	0.52	-1.03 [-1.23, -0.83]
SBP	random segments	GBM, PPG features	14.5	-0.0	18.5	0.43	0.58	-1.66 [-1.92, -1.40]
SBP	random segments	GBM, PPG + demogr.	12.1	-0.0	15.6	0.67	0.58	-4.13 [-4.52, -3.76]
SBP	random segments	1-D CNN	13.1	-0.3	16.7	0.57	0.72	-3.07 [-3.45, -2.69]
SBP	subject-wise	Null (train mean; case mean if personal)	16.2	-0.0	20.4	-0.05	–	–
SBP	subject-wise	1-NN copy probe	18.0	-0.6	22.7	0.15	0.16	+1.74 [+1.47, +2.02]
SBP	subject-wise	Ridge, demographics	16.2	0.0	20.4	-0.00	–	+0.02 [-0.04, +0.07]
SBP	subject-wise	Ridge, PPG features	15.3	0.0	19.3	0.32	0.53	-0.90 [-1.11, -0.69]
SBP	subject-wise	Ridge, PPG + demogr.	15.3	0.1	19.3	0.33	0.53	-0.95 [-1.16, -0.74]
SBP	subject-wise	GBM, PPG features	14.9	0.1	18.9	0.38	0.56	-1.35 [-1.63, -1.07]
SBP	subject-wise	GBM, PPG + demogr.	15.3	0.3	19.4	0.33	0.57	-0.93 [-1.30, -0.56]
SBP	subject-wise	1-D CNN	13.8	-0.3	17.5	0.52	0.69	-2.44 [-2.84, -2.07]
SBP	personal, absolute	Null (train mean; case mean if personal)	13.8	-2.3	18.2	0.47	–	–
SBP	personal, absolute	1-NN copy probe	18.0	-2.2	22.9	0.15	0.15	+4.24 [+3.65, +4.83]
SBP	personal, absolute	Ridge, demographics	15.9	-2.3	20.2	0.04	–	+2.08 [+1.48, +2.69]
SBP	personal, absolute	Ridge, PPG features	15.1	-1.0	19.2	0.31	0.52	+1.26 [+0.64, +1.91]
SBP	personal, absolute	Ridge, PPG + demogr.	15.0	-1.0	19.1	0.32	0.52	+1.18 [+0.58, +1.81]
SBP	personal, absolute	GBM, PPG + demogr.	12.8	-0.9	16.6	0.57	0.53	-0.99 [-1.44, -0.56]
SBP	personal, absolute	1-D CNN	13.7	-2.1	17.5	0.50	0.63	-0.08 [-0.67, +0.52]
SBP	personal, residual	Null (train mean; case mean if personal)	13.8	-2.3	18.2	0.47	–	–
SBP	personal, residual	Ridge, PPG features	13.1	-1.2	17.3	0.54	0.52	-0.68 [-0.90, -0.47]
SBP	personal, residual	GBM, PPG features	12.8	-1.0	16.9	0.57	0.52	-0.99 [-1.24, -0.73]
SBP	personal, residual	GBM, PPG + demogr.	12.6	-0.9	16.7	0.58	0.53	-1.24 [-1.49, -1.00]
SBP	personal, residual	1-D CNN	12.4	-1.4	16.3	0.62	0.67	-1.45 [-1.76, -1.14]
DBP	random segments	Null (train mean; case mean if personal)	10.0	-0.0	12.5	-0.02	-0.01	–
DBP	random segments	1-NN copy probe	10.8	-1.0	13.7	0.19	0.18	+0.84 [+0.64, +1.03]
DBP	random segments	Ridge, demographics	9.7	-0.0	12.3	0.20	-0.02	-0.24 [-0.37, -0.11]
DBP	random segments	Ridge, PPG features	9.4	-0.0	11.8	0.33	0.53	-0.54 [-0.69, -0.40]
DBP	random segments	Ridge, PPG + demogr.	9.0	0.0	11.4	0.42	0.54	-0.96 [-1.15, -0.76]
DBP	random segments	GBM, PPG features	9.0	-0.0	11.4	0.42	0.52	-0.97 [-1.16, -0.79]
DBP	random segments	GBM, PPG + demogr.	6.8	-0.0	8.8	0.73	0.53	-3.20 [-3.49, -2.92]
DBP	random segments	1-D CNN	8.0	-0.3	10.2	0.58	0.67	-1.95 [-2.22, -1.70]
DBP	subject-wise	Null (train mean; case mean if personal)	10.0	0.0	12.5	-0.05	–	–
DBP	subject-wise	1-NN copy probe	11.1	-0.9	14.1	0.15	0.17	+1.14 [+0.94, +1.33]
DBP	subject-wise	Ridge, demographics	9.8	0.0	12.3	0.18	–	-0.18 [-0.32, -0.05]
DBP	subject-wise	Ridge, PPG features	9.5	0.0	11.9	0.32	0.53	-0.51 [-0.66, -0.35]
DBP	subject-wise	Ridge, PPG + demogr.	9.1	0.1	11.5	0.40	0.53	-0.88 [-1.08, -0.69]
DBP	subject-wise	GBM, PPG features	9.3	0.1	11.7	0.36	0.51	-0.71 [-0.91, -0.51]
DBP	subject-wise	GBM, PPG + demogr.	9.1	0.3	11.6	0.41	0.52	-0.85 [-1.12, -0.57]
DBP	subject-wise	1-D CNN	8.5	-0.2	10.8	0.51	0.65	-1.47 [-1.76, -1.21]
DBP	personal, absolute	Null (train mean; case mean if personal)	7.4	-1.2	9.7	0.63	–	–
DBP	personal, absolute	1-NN copy probe	11.1	-1.4	14.0	0.16	0.14	+3.71 [+3.35, +4.08]
DBP	personal, absolute	Ridge, demographics	9.7	-1.2	12.2	0.18	–	+2.27 [+1.91, +2.68]
DBP	personal, absolute	Ridge, PPG features	9.3	-0.3	11.7	0.33	0.52	+1.95 [+1.55, +2.37]
DBP	personal, absolute	Ridge, PPG + demogr.	9.0	-0.3	11.4	0.40	0.50	+1.59 [+1.20, +1.98]
DBP	personal, absolute	GBM, PPG + demogr.	7.2	-0.1	9.3	0.67	0.51	-0.19 [-0.45, +0.06]
DBP	personal, absolute	1-D CNN	8.2	-1.1	10.5	0.53	0.61	+0.82 [+0.44, +1.22]
DBP	personal, residual	Null (train mean; case mean if personal)	7.4	-1.2	9.7	0.63	–	–
DBP	personal, residual	Ridge, PPG features	7.0	-0.6	9.2	0.68	0.50	-0.39 [-0.51, -0.27]
DBP	personal, residual	GBM, PPG features	6.9	-0.5	9.1	0.69	0.51	-0.51 [-0.64, -0.36]
DBP	personal, residual	GBM, PPG + demogr.	6.8	-0.4	9.0	0.70	0.51	-0.62 [-0.75, -0.48]
DBP	personal, residual	1-D CNN	6.6	-0.8	8.6	0.73	0.68	-0.78 [-0.94, -0.61]

TABLE S7

RE-AUDIT OF SOLIMAN *ET AL.* (ARXIV:2411.11094), EVERY SEED. MAE AND RMSE IN MG/DL; Δ: PAIRED MAE DIFFERENCE AGAINST THE NULL WITH ITS 95 % SUBJECT-CLUSTER INTERVAL; A: CLARKE ZONE A (%). STATED ELEMENTS REPRODUCED: 0.5–8 HZ BAND-PASS, 30 HZ, GAUSSIAN-NOISE AUGMENTATION (67 TO 268 SAMPLES), SHUFFLED 10-FOLD CV. UNSTATED ELEMENTS RESOLVED AS DESCRIBED IN THE MAIN TEXT.

Protocol	Seed	Model	MAE	RMSE	R^2	A	Δ [95% CI]
A: augment, then split samples	0	Null	14.45	18.65	-0.01	85	–
A: augment, then split samples	0	1-NN copy probe	0.40	3.30	0.97	100	-14.05 [-16.95, -11.74]
A: augment, then split samples	0	CNN–GRU	1.12	3.47	0.97	100	-13.33 [-16.29, -11.05]
A: augment, then split samples	0, noise ×0.5	Null	14.45	18.65	-0.01	85	–
A: augment, then split samples	0, noise ×0.5	1-NN copy probe	0.40	3.30	0.97	100	-14.05 [-16.95, -11.74]
A: augment, then split samples	0, noise ×0.5	CNN–GRU	0.74	2.76	0.98	100	-13.71 [-16.67, -11.43]
A: augment, then split samples	0, noise ×2	Null	14.45	18.65	-0.01	85	–
A: augment, then split samples	0, noise ×2	1-NN copy probe	0.40	3.30	0.97	100	-14.05 [-16.95, -11.74]
A: augment, then split samples	0, noise ×2	CNN–GRU	1.52	3.77	0.96	100	-12.93 [-15.81, -10.69]
A: augment, then split samples	1	Null	14.51	18.70	-0.01	84	–
A: augment, then split samples	1	1-NN copy probe	0.00	0.00	1.00	100	-14.51 [-17.23, -12.24]
A: augment, then split samples	1	CNN–GRU	1.60	6.17	0.89	98	-12.91 [-16.29, -9.66]
A: augment, then split samples	2	Null	14.45	18.64	-0.00	85	–
A: augment, then split samples	2	1-NN copy probe	0.00	0.00	1.00	100	-14.45 [-17.27, -12.15]
A: augment, then split samples	2	CNN–GRU	0.68	0.92	1.00	100	-13.76 [-16.48, -11.46]
B: split recordings, augment train	0	Null	14.60	18.83	-0.03	85	–
B: split recordings, augment train	0	1-NN copy probe	18.25	25.07	-0.82	66	+3.65 [+0.72, +6.36]
B: split recordings, augment train	0	CNN–GRU	15.21	20.12	-0.17	82	+0.60 [-1.44, +2.75]
B: split recordings, augment train	1	Null	14.69	18.99	-0.04	85	–
B: split recordings, augment train	1	1-NN copy probe	18.51	25.22	-0.84	66	+3.81 [+0.83, +6.47]
B: split recordings, augment train	1	CNN–GRU	16.02	21.44	-0.33	78	+1.32 [-0.95, +3.95]
B: split recordings, augment train	2	Null	14.66	18.97	-0.04	85	–
B: split recordings, augment train	2	1-NN copy probe	18.69	25.29	-0.85	66	+4.03 [+1.23, +6.78]
B: split recordings, augment train	2	CNN–GRU	16.39	22.21	-0.43	76	+1.74 [-0.12, +3.81]
C: split subjects, augment train	0	Null	14.52	18.80	-0.02	84	–
C: split subjects, augment train	0	1-NN copy probe	18.22	24.90	-0.79	69	+3.70 [+0.84, +6.50]
C: split subjects, augment train	0	CNN–GRU	16.39	21.76	-0.37	75	+1.87 [+0.10, +3.81]
C: split subjects, augment train	1	Null	14.52	18.80	-0.02	84	–
C: split subjects, augment train	1	1-NN copy probe	17.90	24.70	-0.76	70	+3.37 [+0.65, +5.87]
C: split subjects, augment train	1	CNN–GRU	16.66	23.20	-0.56	75	+2.13 [-0.44, +4.83]
C: split subjects, augment train	2	Null	14.52	18.80	-0.02	84	–
C: split subjects, augment train	2	1-NN copy probe	18.28	24.91	-0.79	69	+3.76 [+0.98, +6.52]
C: split subjects, augment train	2	CNN–GRU	15.41	20.94	-0.27	84	+0.89 [-1.13, +2.90]

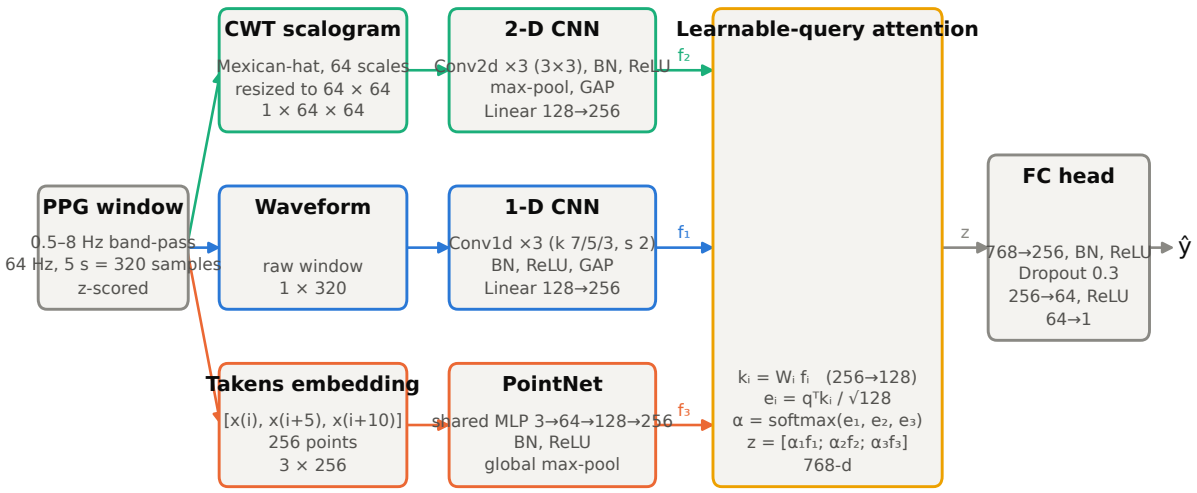


Fig. S1. The 1D–2D–3D fusion network evaluated as one row of the benchmark. One PPG window is presented as a waveform, a wavelet scalogram and a Takens phase-space point cloud; three equal-width encoders feed a learnable-query attention that weights the branches before a regression head.

TABLE S8

ADDITIONAL VALIDATION. DETECTOR STRESS TESTS (100 DRAWS EACH); COPY PROBE ON THE KERMANI–ESMAEILI LADDER; DECISION-RULE FALSE-CLAIM RATE WITH NO SIGNAL (EXACT BINOMIAL 95 % CI); DETECTION RATE OF THE BLOOD-PRESSURE SIGNAL WHEN THE HONEST PREDICTIONS ARE SUBSAMPLED TO GLUCOSE-COHORT SIZES (500 DRAWS; MODELS TRAINED ON THE FULL COHORT); SEED-ENSEMBLE DECISIONS.

Analysis	**Result**
Shuffled cross-join, $k = 3$	sensitivity 0 % (whole column), 0 % (block scan)
Shuffled cross-join, $k = 7$	sensitivity 0 % (whole column), 0 % (block scan)
Shuffled cross-join, $k = 10$	sensitivity 0 % (whole column), 0 % (block scan)
Genuine table stored window-major, 3 records	false alarm 100 % / 100 %; record order: clean
Genuine table stored window-major, 10 records	false alarm 100 % / 100 %; record order: clean
Genuine table stored window-major, 30 records	false alarm 100 % / 100 %; record order: clean
Genuine table stored window-major, 60 records	false alarm 0 % / 0 %; record order: clean
Copy probe (Euclidean), random windows	Δ +0.12 [-2.37, +2.27] mg/dL
Copy probe (shift-invariant), random windows	Δ -2.70 [-5.35, +0.03] mg/dL
Copy probe (Euclidean), random recordings	Δ +0.42 [-1.16, +2.19] mg/dL
Copy probe (shift-invariant), random recordings	Δ -0.44 [-2.50, +1.86] mg/dL
Copy probe (Euclidean), subject-wise	Δ +1.27 [-0.12, +2.76] mg/dL
Copy probe (shift-invariant), subject-wise	Δ -0.51 [-3.03, +1.94] mg/dL
No signal, subject-wise, ridge (1000 cohorts)	false claims 0.4 % [0.1, 1.0]
No signal, subject-wise, gbm (1000 cohorts)	false claims 0.5 % [0.2, 1.2]
No signal, random windows, ridge (1000 cohorts)	false claims 15.8 % [13.6, 18.2]
No signal, random windows, gbm (1000 cohorts)	false claims 99.9 % [99.4, 100.0]
VitalDB SBP, 23 patients $\times$ 3 segments (1-D CNN)	detected in 46 % of draws (mean Δ -2.45)
VitalDB SBP, 16 patients $\times$ all (1-D CNN)	detected in 63 % of draws (mean Δ -2.46)
PPG-BP SBP, 23 subjects (1-D CNN)	detected in 28 % of draws (mean Δ -2.95)
VitalDB DBP, 23 patients $\times$ 3 segments (1-D CNN)	detected in 43 % of draws (mean Δ -1.52)
VitalDB DBP, 16 patients $\times$ all (1-D CNN)	detected in 50 % of draws (mean Δ -1.45)
PPG-BP DBP, 23 subjects (1-D CNN)	detected in 15 % of draws (mean Δ -0.86)
PPG-BP SBP 1-D CNN, seed ensemble of 3	MAE 13.31, Δ -2.94 [-4.37, -1.55]
PPG-BP DBP 1-D CNN, seed ensemble of 3	MAE 7.90, Δ -0.83 [-1.49, -0.16]
Re-audit protocol B, CNN–GRU seed ensemble of 3	MAE 15.44 (null 14.60), Δ +0.84 [-0.80, +2.63]
Re-audit protocol C, CNN–GRU seed ensemble of 3	MAE 15.67 (null 14.52), Δ +1.15 [-0.87, +3.11]

TABLE S9

PUBLISHED RESULTS ON THE KERMANI–ESMAEILI RECORDINGS (23 SUBJECTS, 67 FINGERSTICK REFERENCES) AND THE NULL-ANCHORED RESULT OF THIS PAPER. MAE IN MG/DL.

Study	**Model**	**Split**	**MAE**
Soliman 2024 [4]	CNN–GRU	augment, then shuffle	2.96
Kermani & Esmaeili 2025 [5]	SDF-MFCC + SVR	5-fold, level unstated	4.0†
Syamsul & Kim 2026 [6]	attention CNN–GRU	subject-level (stated)	11.59
Syamsul & Kim 2026 [6]	1-D CNN baseline	subject-level (stated)	13.45
Zeynali 2025 [7]	ResNet-34, VitalDB-trained	external test	14.8
Ramesh 2026 [8]	train-mean null, VitalDB	external test	16.0
This paper	train-mean null	LOSO, recording level	14.7
This paper	1-D CNN, 3 seeds	LOSO, recording level	13.4$\pm$0.5

†reported together with RMSE 26.0 and MARD 0.195%, which are mutually inconsistent.

TABLE S10

LEAKAGE LADDER, 1-D CNN AND NULLS; MAE IN MG/DL. NULL: TRAIN MEAN FOR THE KERMANI–ESMAEILI RECORDINGS AND THE KAGGLE DERIVATIVE; EACH SUBJECT'S OWN TRAINING-PERIOD MEAN FOR BIG IDEAS, WHICH COINCIDES WITH THE LEAKY SUBJECT MEAN. K–E WINDOW, RECORDING AND LOSO RUNGS: THREE SEEDS (MEAN ± STD, THE LOSO RUNG BEING THE EXPERIMENT OF TABLE S11); ALL OTHER RUNGS: ONE SEED. THE BIG IDEAS RANDOM-READINGS RUNG IS SCORED ON ALL 28867 POOLED OUT-OF-FOLD READINGS, THE CHRONOLOGICAL RUNGS ON THE 8664 HELD-OUT READINGS.

Cohort	Protocol	Null	Leaky subj. mean	1-D CNN
K–E	random windows	14.4	12.3	10.4±0.2
	random recordings	14.7	17.8	13.5±0.5
	subject-wise (LOSO)	14.7	–	13.4±0.5
Kaggle deriv.	random windows	9.7	0.03	9.5
	subject-wise (LOSO)	10.2	–	10.2
BIG IDEAs	random readings	15.0	15.0	16.1
	chronological, no gap	15.2	–	16.9
	chronological, 1-h gap	15.3	–	16.9

TABLE S11

LEAVE-ONE-SUBJECT-OUT RESULTS ON THE KERMANI–ESMAEILI RECORDINGS (67 RECORDINGS, 23 SUBJECTS, RECORDING LEVEL). MAE AND RMSE IN MG/DL; A AND A+B ARE CLARKE FRACTIONS IN %. NETWORKS: MAE, RMSE, r AND ZONES ARE MEAN ± STD OVER 3 SEEDS; Δ AND ITS 95% SUBJECT-CLUSTER BOOTSTRAP INTERVAL ARE COMPUTED ON THE SEED-ENSEMBLE PREDICTION (MEAN OF THE SEEDS' PREDICTIONS PER RECORDING), WHOSE MAE IS GIVEN IN PARENTHESES, AGAINST THE TRAIN-MEAN NULL. THE PRIMARY COMPARISON IS MARKED ⋆; OTHER ROWS ARE EXPLORATORY.

	Model	MAE	RMSE	r	A	A+B	Δ vs. null [95% CI]
baseline	Train mean (null)	14.7	18.9	-0.59	83.6	100.0	–
	Reference-noise floor (ISO 15197)	7.1	–	–	–	–	–
	Ridge, demographics	15.6	19.7	-0.17	86.6	100.0	+0.9 [-0.3, +2.3]
	Ridge, PPG features	14.6	18.9	-0.03	86.6	100.0	-0.1 [-0.6, +0.6]
	Gradient boosting, PPG + demogr.	15.5	20.6	-0.05	80.6	100.0	+0.8 [-1.3, +3.0]
network	1-D CNN ⋆	13.4±0.5 (13.3)	18.0±0.2	0.32±0.02	80.6±1.2	100.0±0.0	-1.4 [-3.0, -0.03]
	2-D CNN (CWT)	13.9±0.2 (13.9)	18.5±0.1	0.22±0.01	84.1±0.7	100.0±0.0	-0.8 [-2.5, +0.7]
	PointNet (Takens)	14.0±0.1 (14.0)	18.6±0.1	0.14±0.03	88.1±0.0	100.0±0.0	-0.7 [-1.8, +0.4]
	ResNet-1D [7]	15.1±0.4 (15.0)	19.2±0.5	0.19±0.05	80.6±1.2	100.0±0.0	+0.3 [-1.4, +2.0]
	CNN–LSTM [6]	13.5±0.2 (13.3)	18.7±0.3	0.19±0.02	85.6±0.7	100.0±0.0	-1.4 [-2.9, +0.2]
	1D+2D+3D cross-attention	14.8±0.3 (14.7)	19.3±0.4	0.13±0.05	78.6±0.7	100.0±0.0	-0.1 [-1.5, +1.2]
probe	PaPaGei-S + ridge [9]	14.7	19.1	-0.01	83.6	100.0	+0.0 [-0.9, +1.0]
	Pulse-PPG + ridge [10]	14.2	18.5	0.14	80.6	100.0	-0.5 [-1.7, +0.8]
	AnyPPG + ridge [11]	14.9	19.2	-0.33	85.1	100.0	+0.2 [-0.2, +0.7]

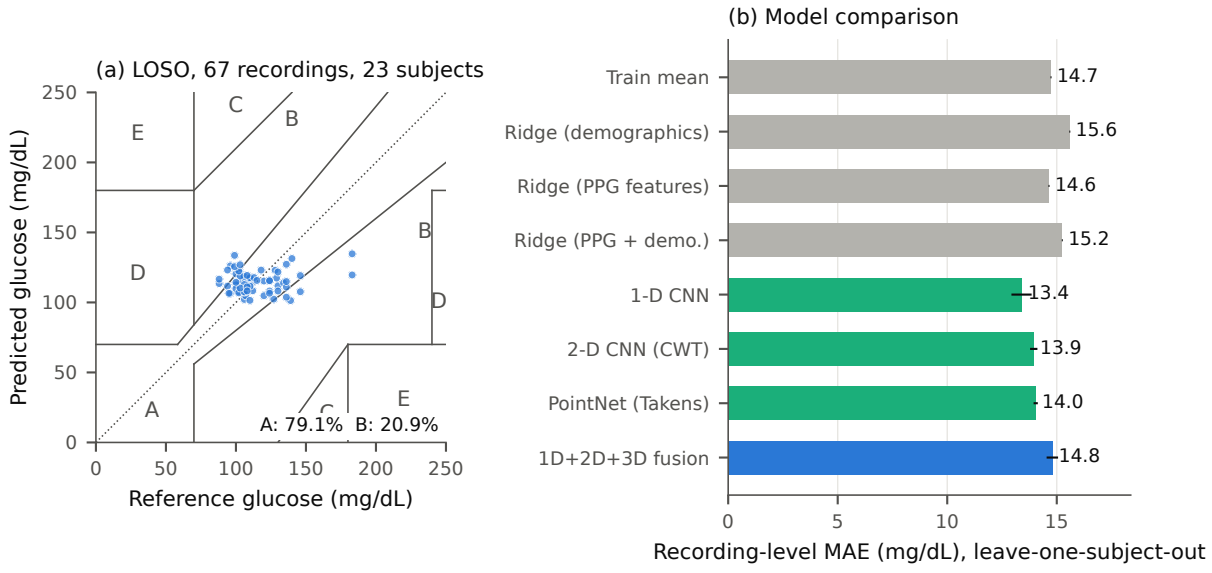


Fig. S2. Leave-one-subject-out results on the Kermani–Esmaeili recordings. (a) Clarke error grid of the fusion network's out-of-fold recording-level predictions: the predictions occupy a narrow band around the training mean. (b) Recording-level MAE of every model; grey bars are baselines that do not see the PPG waveform through a network.

TABLE S12

WITHIN-SUBJECT CHRONOLOGICAL SPLIT WITH 1-H GUARD GAP ON BIG IDEAS (8664 HELD-OUT CGM READINGS, 16 SUBJECTS). MAE AND RMSE IN MG/DL, MARD IN %; r_w: MEDIAN WITHIN-SUBJECT CORRELATION. Δ: PAIRED MAE DIFFERENCE AGAINST THE SUBJECT-MEAN NULL WITH ITS 95% SUBJECT-CLUSTER BOOTSTRAP INTERVAL, COMPUTED ON THE READINGS BOTH ROWS PREDICT. THE ENCODER PROBES EXCLUDE 7 OF 28867 READINGS THAT HAVE NO 10-S SEGMENT PASSING THE QUALITY CHECK, WHICH MOVES THE CHRONOLOGICAL CUT SO THAT 4 OF THE 8664 HELD-OUT READINGS DIFFER (PROBE ROWS ARE SCORED ON 8663 READINGS). 1-D CNN, RESIDUAL: MEAN $\pm$ STD OVER 3 SEEDS; ALL OTHER ROWS ONE RUN.

	Model	MAE	RMSE	MARD	r	r_w	Δ [95% CI]
null	Subject train mean	15.3	21.6	12.8	0.43	–	–
	Reference-noise floor (MARD 9%)	10.5	–	9.0	–	–	–
	Time of day + subject	15.3	21.8	12.8	0.41	-0.01	+0.0 [-0.5, +0.5]
	Persistence	31.3	44.9	27.3	0.21	–	+16.1 [+3.8, +29.9]
population	Ridge, PPG features	17.3	23.8	14.9	0.05	0.05	+2.0 [+0.7, +3.8]
	1-D CNN	16.9	23.8	14.2	0.21	0.04	+1.7 [+0.5, +3.1]
	2-D CNN (CWT)	17.0	23.8	14.3	0.19	0.03	+1.7 [+0.5, +3.2]
	PointNet (Takens)	17.0	23.8	14.3	0.20	0.03	+1.7 [+0.5, +3.2]
	1D+2D+3D cross-attention	16.7	23.5	14.0	0.25	0.03	+1.4 [+0.2, +2.9]
personal.	Ridge, residual	15.1	21.6	12.7	0.44	0.06	-0.1 [-0.4, +0.1]
	1-D CNN, residual	15.0±0.0	21.9±0.0	12.3±0.0	0.44±0.00	0.04±0.00	-0.2 [-0.7, +0.2]
	Fusion, residual	15.1	21.9	12.3	0.44	0.07	-0.2 [-0.7, +0.3]
probe	PaPaGei-S + ridge	17.2	23.8	14.8	0.07	0.03	+2.0 [+0.6, +3.7]
	Pulse-PPG + ridge	17.0	23.6	14.6	0.15	0.04	+1.7 [+0.4, +3.4]
	AnyPPG + ridge	16.8	23.3	14.4	0.23	0.07	+1.5 [+0.3, +3.2]
	PaPaGei-S, residual	15.1	21.6	12.6	0.43	0.03	-0.2 [-0.5, +0.04]
	Pulse-PPG, residual	15.0	21.5	12.5	0.44	0.07	-0.3 [-0.5, -0.02]
	AnyPPG, residual	15.1	21.4	12.6	0.45	0.17	-0.2 [-0.6, +0.2]

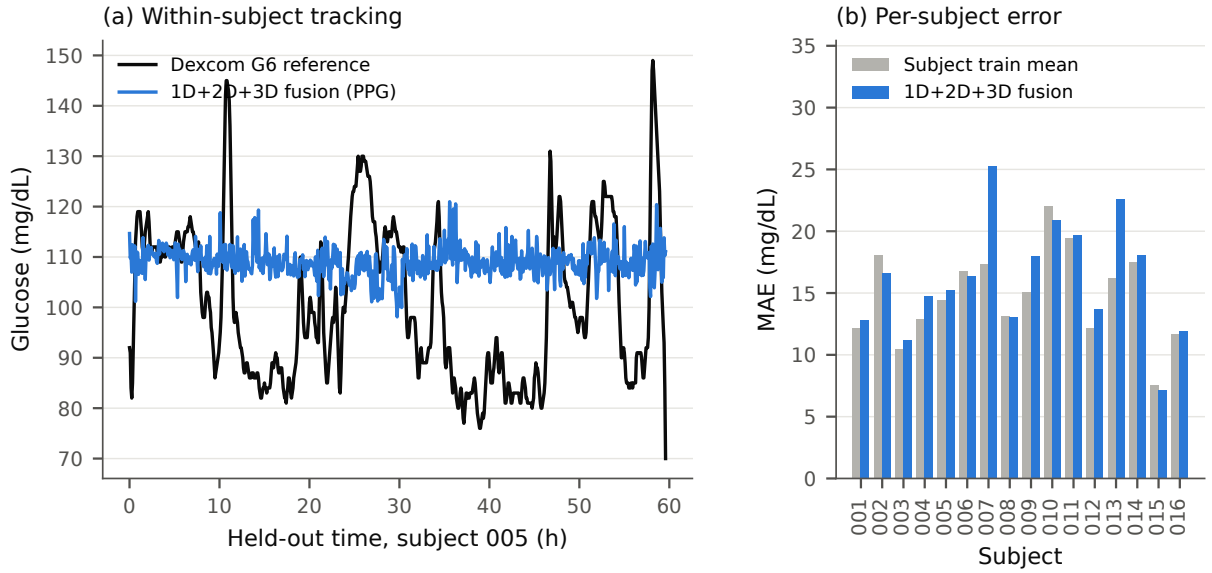


Fig. S3. Within-subject tracking on BIG IDEAs. (a) Dexcom G6 reference and prediction of the population 1D–2D–3D fusion network over the held-out final 30% of one subject's recording. (b) Per-subject MAE of the model against each subject's own training-period mean.

TABLE S13

STUBA EPPG, LEAVE-ONE-SUBJECT-OUT, BY CHANNEL; MAE IN MG/DL. NULL: TRAIN MEAN. 1-D CNN: MEAN $\pm$ STD OVER 3 SEEDS; OTHER CELLS ONE RUN. PAIRED INTERVALS FOR THE IR1 ROWS ARE GIVEN IN TABLE II OF THE MAIN PAPER AND ALL COMPARISONS IN SEC. S11.

Channel	Null	Ridge	GBM	1-D CNN
940 nm, site 1 (IR1)	43.5	44.4	48.3	49.7±1.3
940 nm, site 2 (IR2)	43.5	44.9	45.7	–
655 nm, site 1 (R1)	43.5	43.9	49.3	–
655 nm, site 2 (R2)	43.5	44.9	47.7	–
530 nm, site 1 (G1)	43.5	44.1	47.9	–
530 nm, site 2 (G2)	43.5	44.3	46.8	–
six-channel average	43.5	44.1	46.8	–

TABLE S14

CONTROLS THROUGH THE IDENTICAL PIPELINE AND PROTOCOL (1-D CNN; KERMANI–ESMAEILI ROWS MEAN ± STD OVER THREE SEEDS, BIG IDEAS ROW ONE SEED). MAE IN TARGET UNITS; r: PEARSON.

Cohort	**Target**	**Expected**	**MAE**	**Null**	r
K–E	heart rate (bpm)	recoverable	20.5±0.3	41.9±0.0	0.74±0.01
K–E	age (years)	weak / none	8.9±0.1	8.1±0.0	-0.09±0.01
K–E	glucose	(task)	13.4±0.5	14.7±0.0	0.32±0.02
K–E	glucose, permuted	null	15.6±0.3	14.6±0.0	-0.10±0.13
K–E	glucose, noise input	null	14.6±0.4	14.7±0.0	-0.07±0.09
BIG IDEAs	heart rate (bpm)	recoverable	9.7	14.2	0.63

TABLE S15

REPRESENTATION SENSITIVITY ON BIG IDEAS (GUARDED CHRONOLOGICAL SPLIT, RESIDUAL TARGETS). MAE IN MG/DL; r_w: MEDIAN WITHIN-SUBJECT CORRELATION. ONE WINDOW PER READING AND ONE SEED PER ROW (TABLE S12 USES TWO WINDOWS AND THREE SEEDS, HENCE THE SLIGHTLY DIFFERENT DEFAULT ROW); LONGER WINDOWS KEEP ONLY READINGS WITH A GAP-FREE WINDOW OF THAT LENGTH.

Representation	**Null**	**Ridge**	r_w	**1-D CNN**	r_w
5 s, 0.5–8 Hz, z-score (default)	15.3	15.1	0.04	15.1	0.02
10 s	15.3	15.1	0.05	15.1	0.02
30 s (+ HRV features)	15.3	15.1	0.06	15.0	0.04
60 s (+ HRV features)	15.3	15.1	0.07	15.0	0.08
5 s, 0.1–12 Hz	15.3	15.1	0.03	15.1	0.02
5 s, amplitude retained	15.3	15.1	0.07	15.1	0.04
60 s, 0.1–12 Hz, amplitude, HRV	15.3	15.0	0.10	15.0	0.05

S11. All paired glucose comparisons against the null

Every model-versus-null comparison in the paper (41 in total; $\star$: the pre-specified primary comparison, all others exploratory). MAE of the (seed-ensemble) prediction in mg/dL; Δ: paired MAE difference against the cohort's null (train mean; subject training-period mean on BIG IDEAs) with its 95% subject-cluster bootstrap interval. "better"/"worse": interval entirely below/above zero. 2 intervals lie below zero and 16 above; no correction for multiplicity is applied.

Cohort	Model	MAE	Δ	95% CI	Verdict
Kermani–Esmaeili	Ridge, PPG	14.62	-0.09	[-0.64, +0.62]	–
Kermani–Esmaeili	Ridge, demographics	15.57	+0.86	[-0.32, +2.27]	–
Kermani–Esmaeili	Ridge, PPG + demogr.	15.23	+0.52	[-1.17, +2.57]	–
Kermani–Esmaeili	GBM, PPG	16.61	+1.89	[+0.15, +3.47]	worse
Kermani–Esmaeili	GBM, PPG + demogr.	15.49	+0.78	[-1.28, +2.99]	–
Kermani–Esmaeili	1-D CNN$\star$	13.31	-1.41	[-2.99, -0.03]	better
Kermani–Esmaeili	2-D CNN (CWT)	13.91	-0.81	[-2.51, +0.75]	–
Kermani–Esmaeili	PointNet (Takens)	14.01	-0.70	[-1.83, +0.38]	–
Kermani–Esmaeili	1D+2D+3D fusion	14.66	-0.05	[-1.48, +1.25]	–
Kermani–Esmaeili	ResNet-1D	15.01	+0.30	[-1.42, +2.01]	–
Kermani–Esmaeili	CNN–LSTM	13.34	-1.37	[-2.95, +0.23]	–
Kermani–Esmaeili	PaPaGei-S probe	14.75	+0.03	[-0.91, +1.05]	–
Kermani–Esmaeili	AnyPPG probe	14.91	+0.19	[-0.17, +0.67]	–
Kermani–Esmaeili	Pulse-PPG probe	14.21	-0.50	[-1.67, +0.79]	–
BIG IDEAs	Ridge, PPG	17.32	+2.04	[+0.66, +3.83]	worse
BIG IDEAs	1-D CNN	16.94	+1.66	[+0.48, +3.11]	worse
BIG IDEAs	2-D CNN (CWT)	16.97	+1.69	[+0.50, +3.18]	worse
BIG IDEAs	PointNet (Takens)	16.98	+1.70	[+0.48, +3.22]	worse
BIG IDEAs	1D+2D+3D fusion	16.68	+1.40	[+0.22, +2.85]	worse
BIG IDEAs	Time of day + subject	15.29	+0.01	[-0.49, +0.53]	–
BIG IDEAs	Persistence	31.35	+16.07	[+3.83, +29.92]	worse
BIG IDEAs	Ridge, PPG, residual	15.13	-0.15	[-0.36, +0.09]	–
BIG IDEAs	1-D CNN, residual	15.04	-0.24	[-0.70, +0.23]	–
BIG IDEAs	2-D CNN (CWT), residual	15.09	-0.19	[-0.71, +0.31]	–
BIG IDEAs	PointNet (Takens), residual	15.04	-0.24	[-0.69, +0.17]	–
BIG IDEAs	1D+2D+3D fusion, residual	15.07	-0.21	[-0.73, +0.31]	–
BIG IDEAs	PaPaGei-S probe	17.24	+1.96	[+0.61, +3.68]	worse
BIG IDEAs	AnyPPG probe	16.79	+1.51	[+0.29, +3.20]	worse
BIG IDEAs	Pulse-PPG probe	16.99	+1.70	[+0.38, +3.41]	worse
BIG IDEAs	PaPaGei-S probe, residual	15.07	-0.21	[-0.46, +0.04]	–
BIG IDEAs	AnyPPG probe, residual	15.08	-0.21	[-0.58, +0.16]	–
BIG IDEAs	Pulse-PPG probe, residual	15.01	-0.27	[-0.54, -0.02]	better
STUBA ePPG	Ridge, PPG	44.39	+0.89	[+0.35, +1.22]	worse
STUBA ePPG	GBM, PPG	48.33	+4.83	[+1.76, +9.16]	worse
STUBA ePPG	1-D CNN	49.55	+6.05	[+2.35, +7.65]	worse
STUBA ePPG	2-D CNN (CWT)	48.61	+5.10	[+2.70, +6.32]	worse
STUBA ePPG	PointNet (Takens)	43.20	-0.30	[-1.28, +1.95]	–
STUBA ePPG	1D+2D+3D fusion	47.52	+4.01	[+1.30, +5.16]	worse
STUBA ePPG	PaPaGei-S probe	43.93	+0.43	[-0.01, +1.75]	–
STUBA ePPG	AnyPPG probe	43.51	+0.01	[-0.75, +1.08]	–
STUBA ePPG	Pulse-PPG probe	44.58	+1.08	[+0.07, +1.88]	worse

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